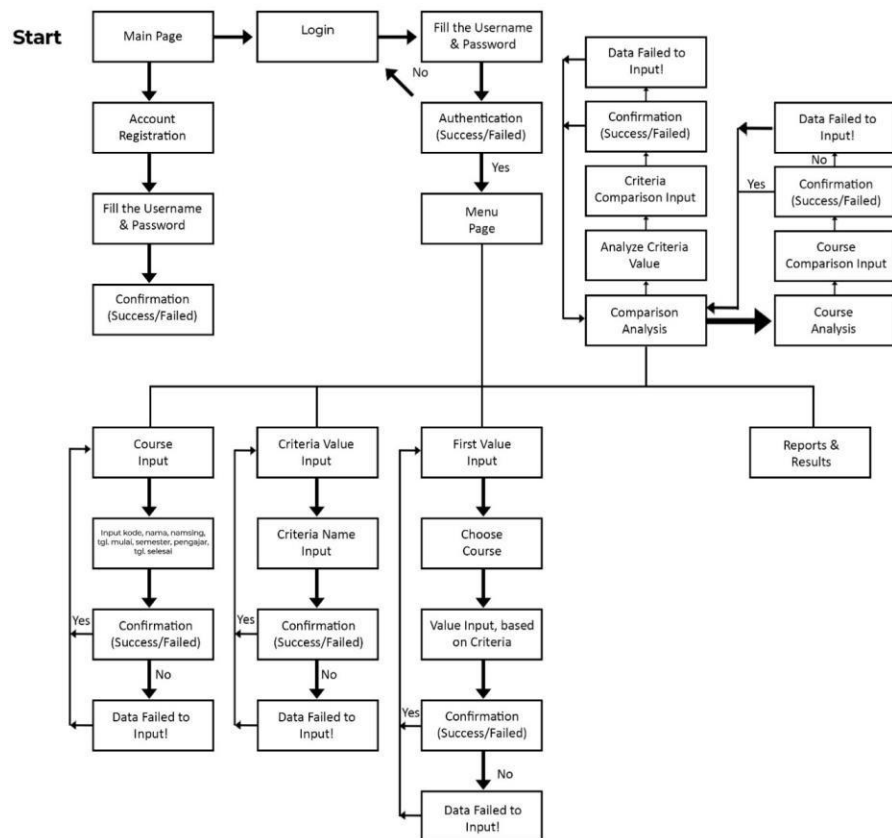


PROFIL BISNIS

INOVASI : SPELL (SISTEM PENILAIAN E-LEARNING)

Rancang bangun SPELL disajikan pada Gambar di bawah ini



Gambar. Desain rancang bangun SPELL

Selanjutnya, dari rancang bangun tersebut disusun pengkodean (koding) dari masing-masing UML. Berikut diberikan contoh tiga koding **(koding lengkap pada setiap rancang bangun terdapat pada dokumen tersendiri)**

Koding Halaman Login

```

<?php
include_once 'includes/config.php';

$config = new Config();
$db = $config->getConnection();
  
```

Back

```
if ($_POST) {
    include_once 'includes/login.inc.php';
    $login = new Login($db);
    $login->userid = $_POST['username'];
    $login->passid = md5($_POST['password']);
    if ($login->login()) {
        echo "<script>location.href='index.php'</script>";
    } else {
        $msg = "Username / Password tidak sesuai!";
    }
}
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>Sistem Penilaian E-Learning</title>
    <link href="images/favicon.ico" rel="icon" type="image/x-icon">
    <link rel="stylesheet" href="assets/css/bootstrap.min.css">
    <link rel="stylesheet" href="assets/css/sweetalert.css">
    <link rel="stylesheet" href="assets/css/style.css">
</head>
<body style="background: #ffffff url(images/back1.jpg) left bottom fixed;">
    <div class="container">
        <div class="row">
            <div class="col-md-4"></div>
            <div class="col-md-4">
```

```

        <form action="<?=$_SERVER['REQUEST_URI']?>"
method="POST">

        <div class="panel panel-dark login-box">

            <div class="panel-body">

                <div class="form-group">

                    <label for="username">Username</label>

                    <input type="text" name="username" class="form-control"
placeholder="Username" autofocus="on">

                </div>

                <div class="form-group">

                    <label for="password">Password</label>

                    <input type="password" name="password" class="form-
control" placeholder="Password">

                </div>

            </div>

            <div class="panel-footer">

                <button type="submit" class="btn btn-dark raised btn-
block">Login</button>

                <div style="text-
align:center">

                    <br>

                    <p>Belum memiliki
akun? <a href="#" target="_blank" style="text-align:center">Daftar di
sini...</a></p>

                </div>

                <br>

                <p class="text-center">Copyright &copy; 2020</p>

                <p class="text-
center">Universitas Muhammadiyah Gresik</p>

            </div>

        </div>

    </form>

```

```

        </div>

        <div class="col-md-4"></div>

    </div>

</div>

<script src="assets/js/jquery-1.11.3.min.js"></script>
<script src="assets/js/sweetalert.min.js"></script>

    <?php if (isset($msg)): ?>
    <script type="text/javascript">

        swal({
            title: "Maaf!",
            text: "<?=$msg?>",
            type: "error",
            timer: 2000,
            confirmButtonColor: "#DD6B55"
        })

    </script>

    <?php endif; ?>
</body>
</html>

```

Koding Data Kriteria

```

<?php
include_once('includes/header.inc.php');
include_once('includes/kriteria.inc.php');

$pro = new Kriteria($db);
$stmt = $pro->readAll();
$count = $pro->countAll();

if (isset($_POST['hapus-contengan'])) {

```

```

$imp = "(" . implode(",", array_values($_POST['checkbox'])) . ")";
$result = $pro->hapusell($imp);
if ($result) { ?>
    <script type="text/javascript">
        window.onload=function(){
            showSuccessToast();
            setTimeout(function(){
                window.location.reload(1);
                history.go(0)
                location.href = location.href
            }, 5000);
        };
    </script> <?php
} else { ?>
    <script type="text/javascript">

```

```

        window.onload=function(){ showErrorToas
            t(); setTimeout(function(){
                window.location.reload(1);
                history.go(0)
                location.href = location.href
            }, 5000);
        };
    </script> <?php
}
?>
<div class="row">
    <div class="col-xs-12 col-sm-12 col-md-12">

```

```

<ol class="breadcrumb">
    <li><a href="index.php">Beranda</a></li>
    <li class="active">Data Kriteria</li>
</ol>

<form method="post">
    <div class="row">
        <div class="col-md-6 text-left">
            <strong style="font-size:18pt;"><span class="fa fa-bank"></span> Data Kriteria</strong>
        </div>
        <div class="col-md-6 text-right">
            <div class="btn-group">
                <button type="submit" name="hapus-contengan" class="btn btn-danger"><span class="fa fa-close"></span> Hapus Contengan</button>

                <button type="button" onclick="location.href='data-kriteria-baru.php'" class="btn btn-primary"><span class="fa fa-clone"></span> Tambah Data</button>
            </div>
        </div>
    </div>
    <br/>
    <table width="100%" class="table table-striped table-bordered" id="tabeldata">
        <thead>
            <tr>
                <th width="10px"><input type="checkbox" name="select-all" id="select-all" /></th>
                <th>ID Kriteria</th>
                <th>Nama Kriteria</th>
                <th>Bobot Kriteria</th>
                <th width="100px">Aksi</th>
            </tr>

```

```

</thead>
<tfoot>
<tr>
<th><input type="checkbox" name="select-all2" id="select-all2" /></th>
<th>ID Kriteria</th>
<th>Nama Kriteria</th>
<th>Bobot Kriteria</th>
<th>Aksi</th>
</tr>
</tfoot>
<tbody>
<?php $no=1; while ($row = $stmt->fetch(PDO::FETCH_ASSOC)) : ?>
<tr>
<td style="vertical-align:middle;"><input type="checkbox" value="<?php
echo $row['id_kriteria'] ?>" name="checkbox[]" /></td>
<td style="vertical-align:middle;"><?php echo $row['id_kriteria'] ?></td>
<td style="vertical-align:middle;"><?php echo $row['nama_kriteria']
?></td>
<td style="vertical-align:middle;"><?php echo $row['bobot_kriteria']
?></td>
<td style="text-align:center;vertical-align:middle;">
<a href="data-kriteria-ubah.php?id=<?php echo $row['id_kriteria'] ?>"
class="btn btn-warning btn-xs"><span class="glyphicon glyphicon-pencil" aria-
hidden="true"></span></a>
<a href="data-kriteria-hapus.php?id=<?php echo $row['id_kriteria'] ?>"
onclick="return confirm('Yakin ingin menghapus data')" class="btn btn-danger
btn-xs"><span class="glyphicon glyphicon-trash" aria-hidden="true"></span></a>
</td>
</tr>
<?php endwhile; ?>
</tbody>
</table>

```

```

</form>

</div>

</div>

<?php include_once('includes/footer.inc.php'); ?>

```

Koding Analisa Kriteria

```

<?php
include_once('includes/header.inc.php');
include_once('includes/kriteria.inc.php');
include_once('includes/nilai.inc.php');

$kriteriaObj = new Kriteria($db);
$nilaiObj = new Nilai($db);

$kriteriaCount = $kriteriaObj->countAll();

$r = [];
$kriterias = $kriteriaObj->readAll();
while ($row = $kriterias->fetch(PDO::FETCH_ASSOC)) {
    $kriteriass = $kriteriaObj->readSatu($row['id_kriteria']);
    while ($roww = $kriteriass->fetch(PDO::FETCH_ASSOC)) {
        $pcs = explode("C", $roww['id_kriteria']);
        $c = $kriteriaCount - $pcs[1];

    }
    if ($c >= 1) {
        $r[$row['id_kriteria']] = $c;
    }
}
?>

<div class="row">
    <div class="col-xs-12 col-sm-12 col-md-12">
        <ol class="breadcrumb">

```



```

</label>Kriteria
Kedua</label>

</div>
</div>
</div>
<?php $no=1; foreach ($r as $k =>
$v): ?>
<?php for ($i=1; $i<=$v;
$i++): ?>
<?php $rows =
$kriteriaObj->readSatu($k); while ($row = $rows-
>fetch(PDO::FETCH_ASSOC)): ?>
<div
class="row">
<div
class="col-xs-12 col-md-3">
<div class="form-group">
<?php $rows = $kriteriaObj->readSatu($k); while($row = $rows-
>fetch(PDO::FETCH_ASSOC)): ?>
<input type="text" class="form-control"
value="<?=$row['nama_kriteria'] ?>" readonly />
<input type="hidden" name="<?=$k?><?=$no?>"
value="<?=$row['id_kriteria']?>" />
<?php endwhile; ?>
</div>
</div>
</div>
<div
class="col-xs-12 col-md-6">

```

```

<div class="form-group">

    <select class="form-control" name="nl<?=$no?>">

        <?php $rows = $nilaiObj->readAll(); while ($row = $rows-
>fetch(PDO::FETCH_ASSOC)): ?>

            <option
value="<?=$row['jum_nilai']?>"><?=$row['jum_nilai']?> -
<?=$row['ket_nilai']?></option>

        <?php endwhile; ?>

    </select>

</div>
</div>
<div
class="col-xs-12 col-md-3">

    <div class="form-group">

        <?php $pcs = explode("C", $k); $nid = "C".($pcs[1]+$i); ?>

        <?php $rows = $kriteriaObj->readSatu($nid); while($row = $rows-
>fetch(PDO::FETCH_ASSOC)): ?>

            <input type="text" class="form-control"
value="<?=$row['nama_kriteria']?>" readonly />

            <input type="hidden" name="<?=$nid?><?=$no?>"
value="<?=$row['id_kriteria']?>" />

```

```

        <?php endwhile; ?>

    </div>

</div>

</div>

<?php endwhile;
$no++; ?>

        <?php endfor; ?>
    <?php endforeach; ?>
    <button type="submit"
name="submit" class="btn btn-dark"> Selanjutnya <span class="fa fa-
arrow-right"></span></button>
        </form>
    </div>
</div>
</div>
</div>
<?php include_once('includes/footer.inc.php'); ?>

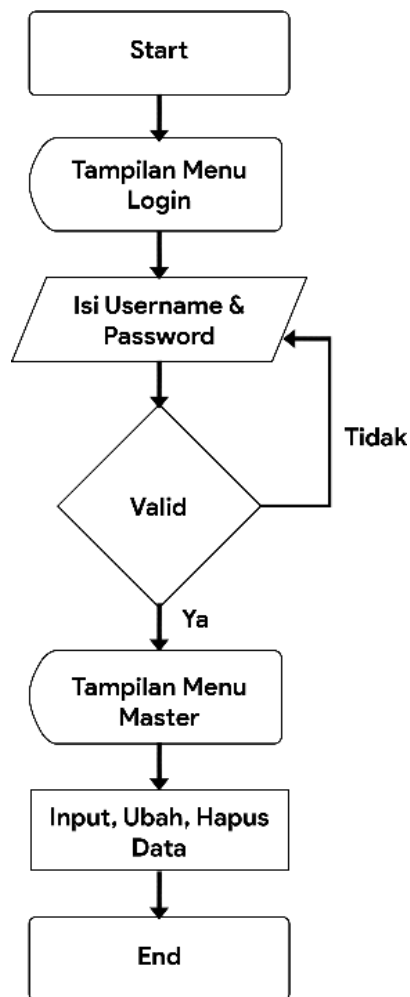
```

Spesifikasi Teknis

Spesifikasi Teknis desain Inovasi SPELL memiliki rancangan antarmuka yang terdiri dari desain 22 halaman, terdiri dari halaman untuk admin, asesor, dan pimpinan institusi.

Secara rinci, dijabarkan kembali sebagai berikut:

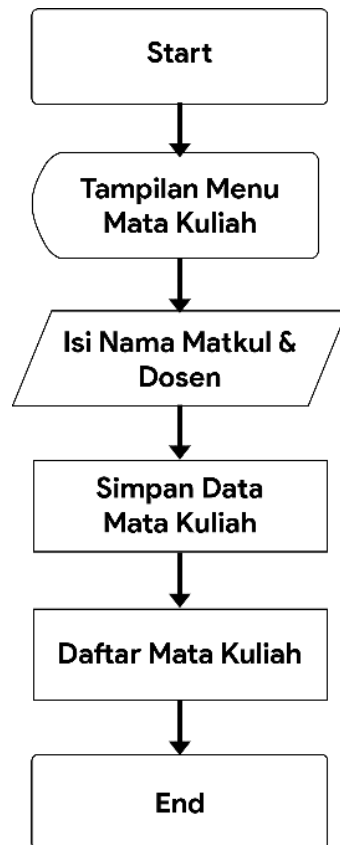
1. Flowchart Login



Gambar 1 Flowchart Login

Flowchart Login menjelaskan bahwa User akan diarahkan ke Halaman Login ketika mengakses website Sistem Penilaian E-Learning. Pada saat itu, User memasukkan Username dan Password yang sudah terdaftar dan dapat diperoleh dari Admin Sistem sendiri. Jika Username dan Password Valid, maka User akan diarahkan ke halaman Menu Master dan dapat melakukan Input, Ubah, dan Hapus Data.

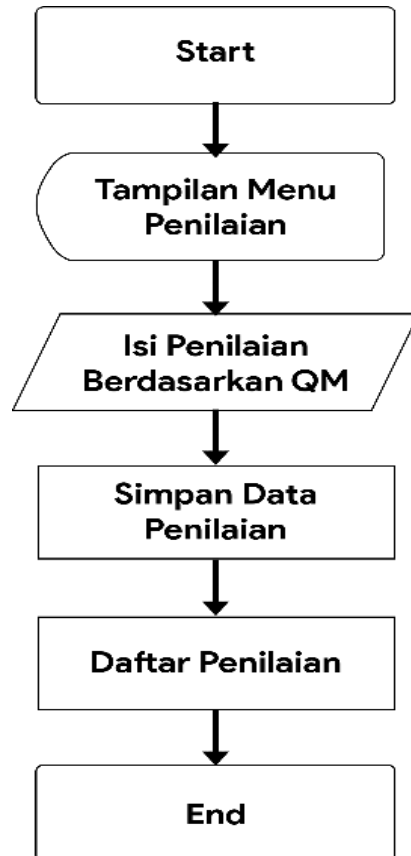
- Flowchart Data Mata Kuliah



Gambar 2 Flowchart Data Input Mata Kuliah

Flowchart Data Mata Kuliah menjelaskan bahwa User berada dalam tampilan menu Mata Kuliah di mana User dapat melakukan Input Mata Kuliah yang diampu. Variabel-Variabel yang digunakan dalam Input Data Mata Kuliah mencakup Nama Mata Kuliah dan Dosen pengampu. Setelah menginputkan data yang diperlukan, maka User dapat melihat Daftar Mata Kuliah yang telah diinputkan.

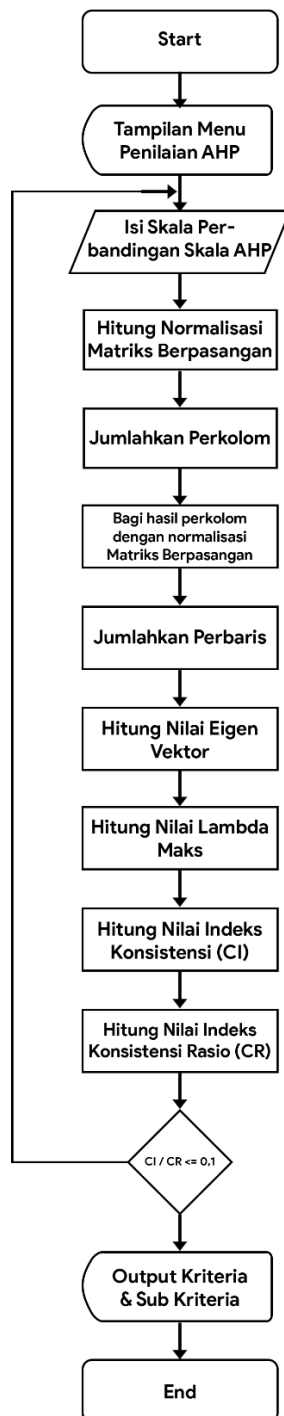
2. Flowchart Penilaian



Gambar 3. Flowchart Penilaian Mata Kuliah berbasis QM

Flowchart Penilaian Mata Kuliah berbasis QM menjelaskan bahwa User akan diarahkan menuju Menu Penilaian. Data Penilaian diisi terlebih dahulu dengan memasukkan penilaian berbasis QM, kemudian memberikan penilaian pada masing-masing Mata Kuliah. Selanjutnya, User dapat melihat Daftar Data Penilaian dari Mata Kuliah yang sudah dinilai.

3. Flowchart Penilaian dan Perbandingan dengan AHP



Gambar 4. Flowchart Perhitungan AHP

Flowchart Perhitungan AHP menjelaskan bahwa User akan diarahkan menuju Menu Penilaian berbasis AHP. Pertama, User akan memasukkan Skala Perbandingan sesuai dengan mekanisme metode AHP, kemudian melakukan

perhitungan normalisasi matriks berpasangan yang akan dijumlahkan per kolom. Setelah selesai melakukan perhitungan akan ditemukan Nilai Eigen Vektor, Nilai Lambda Maks, Nilai Indeks Konsistensi (CI), dan Nilai Indeks Konsistensi Rasio (CR). Jika $CI / CR \leq 0,1$ maka hasilnya konsisten. User dapat melihat Output Data Mata Kuliah dan Penilaian yang sudah dinormalisasi menggunakan AHP serta perangkingan tiap Mata Kuliah.